We are studying the reaction between marble chips (calcium carbonate) and hydrochloric acid. Write a balanced equation for the reaction.

Prediction

Write down what you think will happen and explain why.

Method

1. Pour 25 ml of 1 M hydrochloric acid (HCl) into the conical flask and weigh it. Record the result in the first column of the table.
2. Weigh out 4 g of large marble chips (calcium carbonate). Record the mass in the second column.
3. Add the masses of the reactants together, record the answer in the third column.
4. Place the marble chips into the acid and swirl.
5. When the reaction stops (fizzing stops) re-weigh the flask. Record the mass in the fourth column.
6. Calculate the mass difference in the last column.

Results

Copy and complete the results table.

Mass of acid and flask (g)	Mass of marble chips (g)	Total start mass of reactants (g) **A**	End mass of products (g) **B**	Mass difference (g) **A − B**

Questions

1. Explain any observed changes in mass during the chemical reaction.
2. Explain these changes in terms of the particle model.

Conclusion

1. Describe your results - what have you found out? Is it what you expected?